SQL PROJECT: HOSPITAL RECORDS

Database Analysis & Question Showcase

1 Total Number of Patients

Find the total number of patients across all hospitals.

2 Avg. Doctors Per Hospital

Retrieve the average count of doctors available in each hospital.

3 Top 3 Departments

Find the top 3 hospital departments with the highest number of patients.

4 Max Expenses Hospital

Identify the hospital that recorded the highest medical expenses.

5 Daily Avg. Expenses

Calculate the average medical expenses per day for each hospital.

6 Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

7 Patients Per City

Count the total number of patients treated in each city.

8 Avg. Stay Per Department

Calculate the average number of days patients spend in each department.

9 Lowest Patient Dept.

Identify the department with the least number of patients.

10 Monthly Expenses Report

Group the data by month and calculate the total medical expenses for each month.

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SQL Project_Hospitals_Records/postgres@PostgreSQL 18

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SETUP & VERIFICATION

Task: Verify Hospitals Table Schema & Ingestion

Query | Query History

```
9   Patients_Count INT,
10  Admission_Date DATE,
11  Discharge_Date DATE,
12  Medical_Expenses NUMERIC(10,2)
13
14  );
15
16  Select * from hospitals;
17
18  ---IMPORTING DATA---
19  COPY hospitals(Hospital_name,Location,Department,Doctors_Count,Patients_Count,Admission_Date,Discharge_Date,Medical_Expenses)
20  FROM 'C:\Users\Public\Downloads\Hospital_Data.csv'
21  CSV HEADER;
22
23  --1 Total Number of Patients
```

Data Output | Messages | Notifications

Showing rows: 1 to 100 | Page No: 1 of 1

	record_id [PK] integer	hospital_name character varying (100)	location character varying (50)	department character varying (50)	doctors_count integer	patients_count integer	admission_date date	discharge_date date	medical_expenses numeric (10,2)
1	1	Wellness Clinic	Chennai	Orthopedics	47	182	2023-12-13	2023-12-14	31364.88
2	2	Fortis Care	Pune	ENT	15	51	2023-12-29	2024-01-09	47280.19
3	3	Wellness Clinic	Ahmedabad	Pediatrics	20	120	2023-10-07	2023-10-21	28574.72
4	4	Heritage Hospital	Hyderabad	Urology	8	172	2023-04-29	2023-05-11	7000.83
5	5	City Hospital	Kolkata	Gynecology	35	76	2023-02-10	2023-02-12	47210.46
6	6	Heritage Hospital	Hyderabad	Oncology	11	76	2023-02-05	2023-02-17	18612.34
7	7	Global Medicare	Ahmedabad	Oncology	22	99	2023-08-01	2023-08-06	47808.55
8	8	Apollo Health	Jaipur	General Medicine	37				

Successfully run. Total query runtime: 183 msec. 100 rows affected.

Total rows: 100 | Query complete 00:00:00.186 | CRLF | Ln 17, Col 1

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QUERY 1

Task: Find the total number of patients across all hospitals

Query | Query History

```
18  ---IMPORTING DATA---
19  COPY hospitals(Hospital_name,Location,Department,Doctors_Count,Patients_Count,Admission_Date,Discharge_Date,Medical_Expenses)
20  FROM 'C:\Users\Public\Downloads\Hospital_Data.csv'
21  CSV HEADER;
22
23  --1.Total Number of Patients
24  Select sum(patients_count) AS total_patients
25  From hospitals;
26
27  --2.Average Number of Doctors per Hospital
28  SELECT Hospital_name,location, ROUND(AVG(Doctors_Count), 0) AS Avg_Doctors
29  FROM Hospitals
30  GROUP BY Hospital_name,location
31  ORDER BY Avg_Doctors DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	total_patients bigint
1	9347

Successfully run. Total query runtime: 147 msec. 1 rows affected.

Total rows: 1 Query complete 00:00:00.147 CRLF Ln 25, Col 16

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QUERY 2

Task: Retrieve the average count of doctors available in each hospital

Query | Query History

```
26
27  --2.Average Number of Doctors per Hospital
28  SELECT Hospital_name,location, ROUND(AVG(Doctors_Count), 0) AS Avg_Doctors
29  FROM Hospitals
30  GROUP BY Hospital_name,location
31  ORDER BY Avg_Doctors DESC;
32
33  ---3.Top 3 Departments with the Highest Number of Patients
34  SELECT Department, SUM(Patients_Count) AS Total_Patients
35  FROM Hospitals
36  GROUP BY Department
37  ORDER BY Total_Patients DESC
38  LIMIT 3;
39
40  ---4 Hospital with the Maximum Medical Expenses
```

Data Output | Messages | Notifications

Showing rows: 1 to 65 Page No: 1 of 1

	hospital_name character varying (100)	location character varying (50)	avg_doctors numeric
1	Sunrise Medical	Ahmedabad	48
2	Wellness Clinic	Chennai	47
3	Wellness Clinic	Jaipur	47
4	Healing Touch	Delhi	46
5	Metro Hospital	Hyderabad	46
6	Fortis Care	Mumbai	46
7	Global Medicare	Bangalore	42
8	Green Valley Hospital	Jaipur	42

Total rows: 65 Query complete 00:00:00.226 CRLF Ln 31, Col 27

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Query | Query History

```
26
27  --2.Average Number of Doctors per Hospital
28  SELECT Hospital_name,location, ROUND(AVG(Doctors_Count), 0) AS Avg_Doctors
29  FROM Hospitals
30  GROUP BY Hospital_name,location
31  ORDER BY Avg_Doctors DESC;
32
33  ---3.Top 3 Departments with the Highest Number of Patients
34  SELECT Department, SUM(Patients_Count) AS Total_Patients
35  FROM Hospitals
36  GROUP BY Department
37  ORDER BY Total_Patients DESC
38  LIMIT 3;
39
40  ---4 Hospital with the Maximum Medical Expenses
```

QUERY 3

Task: Find the top 3 hospital departments with the highest number of patients

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	department character varying (50)	total_patients bigint
1	Urology	1766
2	Neurology	1229
3	ENT	1064

Total rows: 3 | Query complete 00:00:00.199 | CRLF | Ln 38, Col 9

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QUERY 4

Task: Identify the hospital that recorded the highest medical expenses

Query | Query History

```
35  FROM Hospitals
36  GROUP BY Department
37  ORDER BY Total_Patients DESC
38  LIMIT 3;
39
40  ---4.Hospital with the Maximum Medical Expenses
41  Select hospital_name, sum(medical_expenses) AS total_expenses
42  From hospitals
43  GROUP BY hospital_name
44  order by total_expenses DESC LIMIT 1;
45
46  ---5.Daily Average Medical Expenses
47  SELECT Hospital_name,
48        ROUND(SUM(Medical_Expenses) / NULLIF(SUM(Discharge_Date - Admission_Date), 0), 2)
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	hospital_name character varying (100)	total_expenses numeric
1	Healing Touch	352178.06

Total rows: 1 | Query complete 00:00:00.189 | CRLF | Ln 44, Col 38

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No limit

Query | Query History

```
43  GROUP BY hospital_name
44  order by total_expenses DESC LIMIT 1;
45
46  ---5.Daily Average Medical Expenses
47  SELECT Hospital_name,
48         ROUND(SUM(Medical_Expenses) / NULLIF(SUM(Discharge_Date - Admission_Date), 0), 2)
49             AS Avg_Expense_Per_Day
50  FROM Hospitals
51  GROUP BY Hospital_name
52  ORDER BY Avg_Expense_Per_Day DESC;
53
54  ---6.Longest Hospital Stay
55  select record_id,hospital_name,location,department,doctors_count,patients_count,admission_date,discharge_date,(discharge_date-admission_
56  FROM hospitals
57  order by stay_days DESC LIMIT 1;
```

QUERY 5

Task: Calculate the average medical expenses per day for each hospital

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	hospital_name character varying (100)	avg_expense_per_day numeric
1	Fortis Care	4385.42
2	Sunrise Medical	4315.76
3	City Hospital	4296.15
4	Global Medicare	3507.87
5	Healing Touch	3486.91
6	Metro Hospital	3245.22
7	Apollo Health	3187.76
8	Wellness Clinic	2830.13

Successfully run. Total query runtime: 305 msec. 10 rows affected.

Total rows: 10 | Query complete 00:00:00.305 | CRLF | Ln 53, Col 1

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SQL Project_Hospitals_Records/postgres@PostgreSQL 18

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QUERY 6

Task: Find the patient with the longest hospital stay (Discharge - Admission Date)

Query | Query History

```
52  ORDER BY Avg_Expense_Per_Day DESC;
53
54  ---6.Longest Hospital Stay
55  select record_id,hospital_name,location,department,doctors_count,patients_count,admission_date,discharge_date,(discharge_date-admission_
56  FROM hospitals
57  order by stay_days DESC LIMIT 1;
58
59  ---7.Total Patients Treated Per City
60  SELECT Location AS City, SUM(Patients_Count) AS Total_Patients
61  FROM Hospitals
62  GROUP BY Location
63  ORDER BY Total_Patients DESC;
64
65  ---8.Average Length of Stay Per Department
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	record_id [PK] integer	hospital_name character varying (100)	location character varying (50)	department character varying (50)	doctors_count integer	patients_count integer	admission_date date	discharge_date date	stay_days integer
1	27	Apollo Health	Lucknow	ENT	9	171	2023-04-06	2023-04-21	15

Successfully run. Total query runtime: 159 msec. 1 rows affected.

Total rows: 1 | Query complete 00:00:00.162 | CRLF | Ln 58, Col 1

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SQL Project_Hospitals_Records/postgres@PostgreSQL 18

No limit

Query | Query History

```
52  ORDER BY Avg_Expense_Per_Day DESC;
53
54  ---6.Longest Hospital Stay
55  select record_id,hospital_name,location,department,doctors_count,patients_count,admission_date,discharge_date,(discharge_date-admission_
56  FROM hospitals
57  order by stay_days DESC LIMIT 1;
58
59  ---7.Total Patients Treated Per City
60  SELECT Location AS City, SUM(Patients_Count) AS Total_Patients
61  FROM Hospitals
62  GROUP BY Location
63  ORDER BY Total_Patients DESC;
64
65  ---8.Average Length of Stay Per Department
```

QUERY 7

Task: Count the total number of patients treated in each city

Data Output | Messages | Notifications

Showing rows: 1 to 10 Page No: 1 of 1

	city character varying (50)	total_patients bigint
1	Jaipur	1505
2	Ahmedabad	1467
3	Lucknow	1264
4	Hyderabad	1261
5	Bangalore	955
6	Chennai	840
7	Kolkata	596
8	Pune	555

Successfully run. Total query runtime: 162 msec. 10 rows affected.

Total rows: 10 Query complete 00:00:00.162 CRLF Ln 63, Col 30

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SQL Project_Hospitals_Records/postgres@PostgreSQL 18

No limit

QUERY 8

Task: Calculate the average length of stay (days) per department

Query | Query History

```
60  SELECT Location AS City, SUM(Patients_Count) AS Total_Patients
61  FROM Hospitals
62  GROUP BY Location
63  ORDER BY Total_Patients DESC;
64
65  ---8.Average Length of Stay Per Department
66  SELECT Department,
67         ROUND(AVG(Discharge_Date - Admission_Date), 2) AS Avg_Stay_Days
68  FROM Hospitals
69  GROUP BY Department
70  ORDER BY Avg_Stay_Days DESC;
71
72  ---9.Identify the Department with the Lowest Number of Patients
73  SELECT Department, SUM(Patients_Count) AS Total_Patients
74  FROM Hospitals
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	department character varying (50)	avg_stay_days numeric
1	Neurology	9.25
2	Pediatrics	9.11
3	Urology	8.72
4	Oncology	8.11
5	ENT	8.08
6	Gynecology	7.67
7	General Medicine	7.43
8	Orthopedics	7.14

Successfully run. Total query runtime: 193 msec. 10 rows affected.

Total rows: 10 | Query complete 00:00:00.193 | CRLF | Ln 70, Col 29

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SQL Project_Hospitals_Records/postgres@PostgreSQL 18

No limit

QUERY 9

Task: Identify the department with the least number of patients

Query | Query History

```
69  GROUP BY Department
70  ORDER BY Avg_Stay_Days DESC;
71
72  ---9.Identify the Department with the Lowest Number of Patients
73  SELECT Department, SUM(Patients_Count) AS Total_Patients
74  FROM Hospitals
75  GROUP BY Department
76  ORDER BY Total_Patients ASC
77  LIMIT 1;
78
79  ---10.Monthly Medical Expenses Report
80  SELECT TO_CHAR(Admission_Date, 'YYYY-MM-DD') AS Month,
81         SUM(Medical_Expenses) AS Total_Expenses
82  FROM Hospitals
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	department character varying (50)	total_patients bigint
1	Cardiology	544

Successfully run. Total query runtime: 167 msec. 1 rows affected.

Total rows: 1 | Query complete 00:00:00.169 | CRLF | Ln 78, Col 1

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No limit

QUERY 10

Task: Group data by month and calculate total medical expenses for each month

Query | Query History

```
73  SELECT Department, SUM(Patients_Count) AS Total_Patients
74  FROM Hospitals
75  GROUP BY Department
76  ORDER BY Total_Patients ASC
77  LIMIT 1;
78
79  ---10.Monthly Medical Expenses Report
80  SELECT TO_CHAR(Admission_Date, 'YYYY-MM-DD') AS Month,
81         SUM(Medical_Expenses) AS Total_Expenses
82  FROM Hospitals
83  GROUP BY TO_CHAR(Admission_Date, 'YYYY-MM-DD')
84  ORDER BY Month;
85
86
```

Data Output | Messages | Notifications

Showing rows: 1 to 86 Page No: 1 of 1

	month text	total_expenses numeric
1	2023-01-11	22456.61
2	2023-01-16	17999.48
3	2023-01-18	82312.75
4	2023-01-20	26646.52
5	2023-01-27	24556.18
6	2023-02-04	42883.43
7	2023-02-05	18612.34
8	2023-02-07	26027.37

Successfully run. Total query runtime: 119 msec. 86 rows affect

Total rows: 86 Query complete 00:00:00.119 CRLF Ln 84, Col 16